TABLE 2
COMPARISON OF DIFFERENT TRANSMITTER AND RECEIVER MODULES

Model	Manufacturer	Transmitter/Receiver	Integrated Antenna	Frequency	Max Power	Qi Compatible
ML7660	Lapis Technology	Receiver	Yes	13.56MHz	1W	No
ML7661	Lapis Technology	Transmitter	Yes	13.56MHz	1W	No
BP3621	ROHM Semiconductor	Transmitter/Receiver	Yes	13.56MHz	240mW	No
BP3622	ROHM Semiconductor	Receiver	Yes	13.56MHz	240mW	No
BQ510103b	Texas Instruments	Receiver	-	110kHz to 205kHz	15W	WPC Qi v1.2
BQ50012A	Texas Instruments	Transmitter/Receiver	-	110kHz to 205kHz	15W	WPC Qi v1.2
MWCT1x23	NXP	Transmitter/Receiver	-	168MHz	65W	WPC Qi v1.2
P9235A-RB	Renesas	Receiver	-	110kHz to 205kHz	5W	WPC Qi v1.2
STWLC98	ST Microelectronics	Receiver/Transmitter	-	110kHz to 205kHz	70W	Qi 1.2, Qi 1.3
STWBC2	ST Microelectronics	Transmitter	-	110kHz to 205kHz	70W	Qi 1.2, Qi 1.3

Higher power transfer rate

Although the overall efficiency of portable devices, such as smartphones and smartwatches, is improving, the higher processing power and overall improvement in features mean that these devices are getting more power-intensive, thus requiring bigger batteries. To improve the battery charging speed, manufacturers have been trying to increase the power transfer rate by wireless charging.

STM STWLC98 is a Qi-compliant 70W inductive power receiver that can also act as a 15W wireless transmitter. The NXP MWCT1123 is a power transmitter controller module based on the WPC standard that can transmit power up to 65W.

Better protection features

Newer wireless transmitters and receivers are featuring a lot of functionality and are capable of transferring higher power, which makes them susceptible to cause unwanted problems like overheating, transferring overvoltage, etc. Therefore, the latest modules come with various protection features to ensure the reliability of the device.

TI BQ51013B-Q1 and STM STWLC98 feature foreign object detection, overcurrent and overvoltage protection, thermal shutdown, and fault host control. Some modules, such as NXP's MWCT1123, feature a memory protection unit whereas some devices, such as STM STSAFE-V110, provide cryptographical security.

More number of interfaces

With the inclusion of new features and powerful processors in the wireless charging modules it can be safely said that the new devices come with an increased number of interfaces.

For communication, the Lapis ML7660/7661, NXP MWCT1123, and STM STWBC2 use SPI/I2C master UART as a serial interface. The Renesas P9235A features five GPIOs while STM's STWBC2 has eight GPIOs for connecting with external devices.

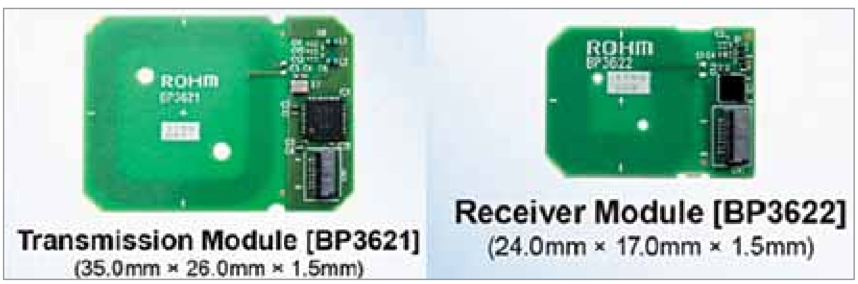

Fig. 3: ROHM BP3621 and BP3622 wireless charging modules (Source: Datasheet)

Smaller modules for smarter wearables

Electronic gadgets are becoming more powerful and wireless at the same time, be it your smartwatch, wireless earphones, or any other gadget. Moreover, the increase in penetration of IoT in our everyday lives demands compact wireless charging modules. Companies such as TI, ROHM, and Lapis are competing to create compact modules.

Lapis ML7660/ML7661 is said to be the smallest wireless receiver capable of delivering 1W power while ROHM BP3621 and BP3622 wireless charging modules come with an integrated antenna and feature a small form factor, thus reducing the size of product along with time and cost of development.

However, there are many more upcoming tends. Some of these are described below.

GaN replacing silicon FET

Adoption of GaN transistors is the next big thing in wireless, which will improve the range and power transmission rate. GaN can help charge at rates from 30W up to several kilowatts, which was difficult to achieve using the silicon based charger. GaN based chargers are enabling higher charge transfer rates and better efficiency at a lower cost compared to silicon based solutions. The GaN based